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ANALYSIS OF AN HIV/AIDS MODEL WITH PUBLIC-HEALTH INFORMATION CAMPAIGNS AND INDIVIDUAL WITHDRAWAL

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Primary prevention measures designed to alter susceptibility and/or reduce exposure of susceptible individuals to diseases, remain the mainstay in the fight against HIV/AIDS. A model for HIV/AIDS, that investigates the reduction in infection by advocating for sexual behavior change through public-health information campaigns and withdrawal of individuals with AIDS from sexual activity is proposed and analyzed. The contact rate is modeled using an incidence function with saturation that depends on the number of infectives. The dynamics of the model is determined using the model reproduction number $\mathcal{R}_0$. Numerical simulations are presented to illustrate the role of some key epidemiological parameters. The results from the study demonstrate that an increase in the rate of dissemination of effective public-health information campaigns results in a decrease in the prevalence of the disease. Similarly, an increase in the fraction of individuals with AIDS who withdraw from sexual activities reduces the burden of the disease.

Keywords: Information Campaigns; HIV/AIDS; Stability; Reproduction Number; Contact Rate.

1. Introduction

HIV/AIDS remains a global challenge since the identification of the first cases in 1981.[1] It has caused huge demographic changes in most of the heavily affected countries. Sub-Saharan Africa hosts the majority of people living with HIV/AIDS, an estimated 67% of the total global cases.[2–4] In 2007, there were an estimated 2.7

million HIV new infections globally and 35% of them were from Southern Africa. Currently, the epidemic appears to have stabilized at endemic levels, especially in Southern Africa. Reductions in HIV prevalence have been observed in Botswana and Zimbabwe. The global consensus currently is that, prevention of new infections is the key to reversing the HIV/AIDS epidemic. While 87% of countries with targets for universal access to treatment have established goals for HIV treatment, sadly only half of them have targets for key HIV prevention strategies.[4] While effective strategies exist to prevent every mode of HIV transmission, prevention efforts are most effective when their impact is known and has been quantified. Accelerated progress towards reversing the HIV/AIDS epidemic can be achieved by using evidence-informed HIV prevention strategies. HIV primary prevention strategies include behavior change, correct and consistent use of condoms, reduction in the number of sexual partners and media campaigns.[2,4]

Advances in HIV education and prevention and their impact on the HIV epidemic have been investigated by many authors, see for instance Refs. 5–8. Mathematical models have also been used to investigate the impact of HIV prevention on the epidemic.[2,3,9–11] The impact of media coverage on infectious diseases has recently been modeled.[12,13] Public awareness has been used, both as a primary prevention and post-exposure risk reduction strategy.[14,15] Prevention that promotes attitudinal and behavioral change, low-risk sexual behavior, has been recommended as the only effective strategy to control HIV infection. Media coverage and information dissemination has been unprecedented in its effects on behavior change in Uganda, a country reported to have had the greatest success in combating HIV/AIDS.[2] It is against this background that we use a compartmental model that endeavors to answer the following questions; what are the effects of media campaigns on the long term dynamics of HIV? Can we quantify the effect of media campaigns on the prevalence of HIV? We assume that campaigns reduce the contact rate between susceptible and the infected individuals. We suggest a mathematical function that models the contact rate as a function of infected individuals in the population.

A distinct aspect of epidemic models is the description of the infection process. In the case of heterogenous HIV transmission, infection occurs when an uninfected individual comes in contact with an infected individual sexually. The number of contacts that result in an infection, made between the individuals per unit time, define the contact rate. The contact rate, which determines the rate of transmission of the disease, depends on the transmission probability (the probability that a contact will result in an infection), the number of contacts made and the mixing patterns of the population.[16] Mathematical functions have been used to model the contact rate.[12,13,17] In Ref. 12, the contact rate function is given by $\beta(I) = \mu e^{-mI}$, where I is the number of infectives, μ a constant and m reflects the impact of media coverage. A similar function, $f(H) = de^{-\lambda H}$ is used in Ref. 17 representing the reduced transmission rate of HIV. d is the maximum HIV transmission rate,

H the HIV prevalence and λ is a parameter that measures behavior change in the population. In Ref. 13 the contact rate after a media alert is represented by the expression $\beta_1 - \beta_2 \frac{I}{m+I}$, where β_1 and β_2 are contact rates before and after media alert respectively. In Ref. 18, a sigmoidal function $g(\nu) = \frac{(\nu/m_0)^k}{1+(\nu/m_0)^k}$, $k > 1$, was used to model the infection rate, with $m_0 > 0$ a constant and k the slope of the sigmoidal curve. This function actually accommodates the infection rates $g(I) = \frac{\beta k I}{1+\alpha I}$ in Ref. 19, $g(I) = \frac{\beta I}{1+I}$ in Ref. 20 and $g(I) = \frac{\beta I^2}{1+\alpha I^2}$ in Ref. 21. The terms $1/(1 + \alpha I)$ and $1/(1 + \alpha I^2)$ measure the inhibition effect resulting from behavioral change or the crowding effect of infected individuals,[22] with α measuring the inhibition effect. We revisit the saturation incidence taking into account that there can be more than one infected individual. We are also interested in the proportion q, of individuals who withdraw from sexual activities due to them knowing their HIV status, see for instance Ref. 23 By withdrawal in our context, we mean a voluntary decision by an individual to withdraw from sexual activities as a result of either their knowledge HIV infection status or the disease. Therefore a proportion $(1 - q)$ engage in sexual activities.

This paper is organized as follows: In the next section, the model is formulated and analyzed. It is in this section that we discuss the existence and stability of the model equilibria. Numerical simulations are given in Sec. 3 and the paper ends with a discussion in Sec. 4.

2. Model Formulation and Analysis

We consider a sexually active population $N(t)$, at time t. In the absence of treatment and other post-exposure intervention, an adult individual's survival can be modelled using the four stage model of HIV disease progression, with the four stages corresponding to the WHO Clinical Staging System.[16] Depending on the infection stage, we subdivide the population into subclasses (compartments): susceptibles $S(t)$, asymptomatic infectives $I_1(t)$ (infectious individuals who do not show symptoms of the disease), symptomatic infectives $I_2(t)$ (HIV infected individuals who show symptoms of the disease) and those with full blown AIDS $A(t)$, who are assumed to be active in spreading the infection. The mode of transmission is assumed to be via heterosexual contacts as this represents the single major primary mode of HIV infection globally, especially in the worst affected regions of the world. Our sexually active population is thus given by $N(t) = S(t) + I_1(t) + I_2(t) + A$. Each susceptible individual is considered to be equally likely to be infected by an infectious individual, i.e, the population mixes homogeneously. All parameters of the model are assumed positive. The recruitment rate of susceptible individuals is given by μb and μ is the per capita background mortality. The transfer rate from the asymptomatic compartment to the symptomatic compartment is σ. The removal rate of the symptomatic infectives as they develop AIDS is given by ρ.

The disease-related death rate is given by δ. We thus have the following system of equations;

$$\begin{aligned}\frac{dS}{dt} &= \mu b - \mu S - \lambda(\mathbf{I}, A)S,\\ \frac{dI_1}{dt} &= \lambda(\mathbf{I}, A)S - (\mu+\sigma)I_1,\\ \frac{dI_2}{dt} &= \sigma I_1 - (\mu+\rho)I_2,\\ \frac{dA}{dt} &= \rho I_2 - (\mu+\delta)A,\end{aligned} \tag{2.1}$$

where $\lambda(\mathbf{I}, A) = \frac{c\beta\{I_1+\eta_1 I_2+\eta_2(1-q)A\}}{1+\alpha\{I_1+\eta_1 I_2+\eta_2(1-q)A\}}$, η_1 and η_2 measure the relative infectivity of I_2 and A, when compared to I_1 and $\mathbf{I} = (I_1, I_2)$. We assume that all parameters are positive and the initial conditions of model system (2.1) are given as

$$S(0) = S_0 > 0, \quad I_1(0) = I_{10} > 0, \quad I_2(0) = I_{20} > 0, \quad A(0) = A_0 > 0.$$

The mean number of sexual partners is given by c and β is the per partnership transmission rate. α quantifies effectiveness of information as it spreads in preventing HIV transmission in an environment with public health HIV/AIDS information campaigns.

The proposed contact rate is set to depend on the number of infected individuals in the population. It is determined by $\lambda(\mathbf{I}, A)$ which is of the form $\frac{c\beta g(\mathbf{I},A)}{\psi(\mathbf{I},A)}$, where $g(\mathbf{I}, A) = I_1(t) + \eta_1 I_2(t) + \eta_2(1-q)A$. Here $\psi(\mathbf{I}, A) = 1 + \alpha g(\mathbf{I}, A)$. This incidence was used to generalize the incidence rate $\frac{\beta I^{p_1} S}{1+aI^{q_1}}$ in Ref. 24 but only for a single infective population. Obviously, if $\alpha = 0$, then $\psi(\mathbf{I}, A) = 1$.

Also $\psi(\mathbf{I}, A) \geq 1$ for all $I_1(t) \geq 0$, $I_2(t) \geq 0$ and $A(t) \geq 0$. Many researchers have considered the case $\alpha = 0$, see for instance.[3,10,11,25,26] In essence, we are using a saturation incidence to model HIV/AIDS transmission. The assumption here is that the rate of HIV transmission in the community is basically determined by the amount of HIV/AIDS public health information available in the community. In particular, it is important to note that $\frac{c\beta g(\mathbf{I},A)}{\psi(\mathbf{I},A)}$ tends to a close approximation of the term $c\beta g(\mathbf{I}, A)$. The incidence function thus reduces to a mass action incidence function in this case. However, $\frac{c\beta g(\mathbf{I},A)}{\psi(\mathbf{I},A)}$ approaches the constant $\frac{c\beta}{\alpha}$ for very large values of I_1, I_2 and A.

2.1. *Invariant regions*

Lemma 2.1. *The region $\mathcal{G}$ defined by*

$$\mathcal{G} = \Big\{(S(t), I_1(t), I_2(t), A(t)) \in \mathbb{R}^4_+ : N \leq b\Big\},$$

is positively invariant and attracting with respect to model system (2.1).

Proof. Let $(S(t), I_1(t), I_2(t), A(t))$ be any solution of the system with non-negative initial conditions $(S_0,\ I_{10},\ I_{20},\ A_0)$. Since $\frac{dS}{dt} + (\lambda(\mathbf{I}, A) + \mu)S = \mu b$, it follows that

$$\frac{d}{dt}\left(S(t)e^{\int_0^t (\lambda(\mathbf{I},A)+\mu)ds}\right) \geq 0,$$

$S(t)e^{\int_0^t (\lambda(\mathbf{I},A)+\mu)ds}$ is an increasing function of t, thus S stays positive. The positivity of I_1, I_2 and A can be proved along the same lines as follows.

$$\frac{dI_1}{dt} > -(\sigma + \mu)I_1.$$

Therefore

$$I_1 > I_{10}e^{-(\sigma+\mu)t} \geq 0.$$

Likewise

$$A > A_0 e^{-(\delta+\mu)t} \geq 0.$$

Taking the time derivative of our total population along its solution path gives,

$$\frac{dN}{dt} = \mu b - \mu N - \delta A.$$

We then have

$$\begin{aligned} \frac{dN}{dt} &\leq \mu b - \mu N - \delta A \\ &< \mu b - \mu N. \end{aligned}$$

We note that $\frac{dN}{dt} \leq 0$ if $N \geq b$, since the right hand side is bounded above by $\mu(b - N)$. It can be easily shown that $N(t) \leq N_0 e^{-\mu t} + b(1 - e^{-\mu t})$. Thus $N(t) \leq b$ if $N_0 \leq b$. If $N_0 > b$, then $(S(t), I_1(t), I_2(t), A)$ enters $\mathcal{G}$ or approaches it asymptotically. Hence, the region of biological interest $\mathcal{G}$ is positively invariant and attracting under the flow induced by system (2.1). Furthermore, in $\mathcal{G}$, the usual existence, uniqueness and continuation results hold for model system (2.1), making the model system well-posed mathematically and epidemiologically. Hence, it is sufficient to consider the dynamics of the flow generated by the model (2.1) in $\mathcal{G}$. □

2.2. *Disease-free equilibrium*

It is easy to check that system (2.1) has a disease-free equilibrium given by

$$\mathcal{E}_0 = \Big(b, 0, 0, 0\Big).$$

We can also easily show that $\mathcal{E}_0$ is attracting and its linear stability is governed by the model reproduction number $\mathcal{R}_e$.

2.2.1. $\mathcal{R}_0$ *and the local stability of* $\mathcal{E}_0$

The model reproduction number, $\mathcal{R}_0$, is determined by the decomposition technique in Ref. 27. A reproduction number obtained this way determines the local stability of the disease-free equilibrium point with local asymptotic stability for $\mathcal{R}_0 < 1$ and instability for $\mathcal{R}_0 > 1$.[28] Using the approach in Ref. 27 and adopting the matrix definitions, the matrices for new infection terms and the transfer terms are respectively given by

$$F = \begin{pmatrix} c\beta b & c\beta\eta_1 b & c(1-q)\beta\eta_2 b \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad V = \begin{pmatrix} \mu+\sigma & 0 & 0 \\ -\sigma & \mu+\rho & 0 \\ 0 & -\rho & \mu+\delta \end{pmatrix}.$$

The reproduction number is given by the spectral radius (the dominant eigenvalue) of the matrix FV^{-1}, denoted by $\rho(FV^{-1})$. Thus,

$$\rho(FV^{-1}) = \mathcal{R}_0 = \frac{\beta bc}{\mu+\sigma}\left[1 + \frac{\eta_1\sigma}{\rho+\mu} + \frac{\eta_2\rho\sigma(1-q)}{(\rho+\mu)(\delta+\mu)}\right]. \tag{2.2}$$

Following, Theorem from Ref. 27, we have the following result:

Theorem 2.1. *The disease-free equilibrium of system* (2.1) *is locally asymptotically stable whenever* $\mathcal{R}_0 < 1$ *and unstable otherwise.*

$\mathcal{R}_0$ can be expressed as

$$\mathcal{R}_0 = b\left\{R_{I_1} + \frac{\sigma}{\sigma+\mu}R_{I_2} + \frac{\sigma\rho(1-q)}{(\rho+\mu)(\sigma+\mu)}R_A\right\},$$

where

$$R_{I_1} = \frac{\beta c}{\mu+\sigma}, \quad R_{I_2} = \frac{\eta_1\beta c}{\rho+\mu} \quad \text{and} \quad R_A = \frac{\eta_2\beta c}{\delta+\mu}$$

represent the contribution of the asymptomatic, symptomatic and AIDS individuals to the overall model reproduction number $\mathcal{R}_0$ respectively. The proportions of asymptomatic individuals who become symptomatic and individuals who develop full-blown AIDS are given by $\frac{\sigma}{\mu+\sigma}$ and $(\frac{\rho}{\mu+\rho})(\frac{\sigma}{\mu+\sigma})$ respectively.

2.2.2. *Global stability of* $\mathcal{E}_0$

To prove globally stability of the disease-free equilibrium point, we need the following lemma. For a bounded real-valued function f on $[0,\infty)$, let

$$f_\infty = \lim_{t\to\infty}\inf_{\theta\ge t} f(\theta), \qquad f^\infty = \lim_{t\to\infty}\sup_{\theta\ge t} f(\theta).$$

Lemma 2.2.[29] *Let* $f : [0,\infty) \to \mathbb{R}$ *be bounded and twice differentiable with a bounded second derivative. Let* $t_n \to \infty$ *and* $f(t_n)$ *converges to* f_∞ *or* f^∞ *for* $n \to \infty$ *then* $f'(t_n) \to 0,\ n \to \infty$.

Theorem 2.2. *The disease-free equilibrium $\mathcal{E}_0$ of* (2.1) *is globally asymptotically stable in $\mathcal{G}$ if $\mathcal{R}_0 < 1$.*

Proof. The first equation of (2.1) gives $\frac{dS}{dt} \leq \mu(b - S)$. The solution of $\frac{d\hat{S}}{dt} = \mu(b - \hat{S})$, is a super solution of $S(t)$ i.e $\hat{S}(t) \geq S(t)$ for all $t \geq 0$. Since $\hat{S} \to b$ as $t \to \infty$, then given an $\epsilon > 0$, $\exists,\ t = t_0 \geq 0$ such that

$$S(t) \leq \hat{S}(t) \leq b + \epsilon.$$

The second equation of system (2.1) can be written as

$$\frac{dI_1}{dt} \leq c\beta(b+\epsilon)I_1 - (\mu+\sigma)I_1, \quad \text{for t large enough.} \tag{2.3}$$

Combining (2.3) and the remaining equations of system (2.1) we have the following matrix representation,

$$\begin{pmatrix} \dfrac{dI_1}{dt} \\ \dfrac{dI_2}{dt} \\ \dfrac{dA}{dt} \end{pmatrix} \leq \mathcal{A} \begin{pmatrix} I_1 \\ I_2 \\ A \end{pmatrix}, \tag{2.4}$$

where

$$\mathcal{A} = \begin{pmatrix} c\beta(b+\epsilon) - (\mu+\sigma) & c\beta\eta_1(b+\epsilon) & c\beta\eta_2(1-q)(b+\epsilon) \\ \sigma & -(\mu+\rho) & 0 \\ 0 & \rho & -(\mu+\delta) \end{pmatrix}. \tag{2.5}$$

Let $u \in \mathbb{R}^+$, such that $u > \max\{\mu+\sigma, \mu+\rho, \mu+\delta\}$, we observe that $\mathcal{A} + u\mathcal{I}_{[3]}$, is a nonnegative irreducible matrix with $\mathcal{I}_{[3]}$ being a 3×3 identity matrix. If ω_1, ω_2 and ω_3 are eigenvalues of $\mathcal{A}$, then clearly $\omega_1 + u, \omega_2 + u$ and $\omega_3 + u$ will be the corresponding eigenvalues of $\mathcal{A} + u\mathcal{I}_{[3]}$.

It thus follows from the Perron-Frobenius theorem for non-negative matrices,[30] that $\mathcal{A} + u\mathcal{I}_{[3]}$ has a simple positive eigenvalue given by the spectral radius with a corresponding left eigenvector $\mathbf{v}$ whose components are all positive. In this case all the eigenvalues will be real. Suppose $\omega_1 + u$ is the dominant eigenvalue (the spectral radius) of $\mathcal{A} + u\mathcal{I}_{[3]}$, then $\omega_1 > \omega_2 > \omega_3$ and $\mathbf{v}\mathcal{A} = \omega_1\mathbf{v}$. In this case the eigenvalues are the roots of the characteristic polynomial

$$P(\vartheta) = \vartheta^3 + \hat{a}_2\vartheta^2 + \hat{a}_1\vartheta + \hat{a}_0 = 0, \tag{2.6}$$

where

$$\begin{aligned}
\hat{a}_2 &= (\mu+\rho)+(\mu+\delta)+(\mu+\sigma)\left(1-\frac{c\beta(b+\epsilon)}{\mu+\sigma}\right),\\
&= (\mu+\rho)+(\mu+\delta)+(\mu+\sigma)(1-R^1),\\
\hat{a}_1 &= (\mu+\delta)(\mu+\rho)+(\mu+\sigma)[(\mu+\rho)\\
&\quad +(\mu+\delta)]\left(1-\frac{c\beta(b+\epsilon)}{\mu+\sigma}\left[1+\frac{\eta_1\sigma}{(\mu+\rho)+(\mu+\delta)}\right]\right),\\
&= (\mu+\delta)(\mu+\rho)+(\mu+\sigma)[(\mu+\rho)+(\mu+\delta)](1-R^2),\\
\hat{a}_0 &= (\mu+\rho)(\mu+\sigma)(\mu+\delta)(1-\mathcal{R}),
\end{aligned}$$

where

$$\begin{aligned}
R^1 &= \frac{c\beta(b+\epsilon)}{\mu+\sigma}, \quad R^2 = \frac{c\beta(b+\epsilon)}{\mu+\sigma}\left[1+\frac{\eta_1\sigma}{(\mu+\rho)+(\mu+\delta)}\right],\\
\mathcal{R} &= \frac{c\beta(b+\epsilon)}{\mu+\sigma}\left[1+\frac{\eta_1\sigma}{\rho+\mu}+\frac{\eta_2\rho\sigma(1-q)}{(\rho+\mu)(\delta+\mu)}\right].
\end{aligned}$$

We thus have the following relation $R^1 \le R^2 \le \mathcal{R}$. Note that as $\epsilon \to 0$, $\mathcal{R} \to \mathcal{R}_0$. Thus $\hat{a}_0 > 0$ if ϵ is sufficiently small and by comparison, $\hat{a}_1, \hat{a}_2$ are also positive whenever $\mathcal{R}_0 < 1$. If $\hat{a}_1\hat{a}_2 > \hat{a}_0$, then all the solutions of (2.6) have negative real parts (from the Routh-Hurwitz conditions). This guarantees that all the the eigenvalues have negative real parts. So from (2.4), it follows that for $t \ge t_0$

$$\frac{d}{dt}(\mathbf{v}\cdot[I_1(t), I_2(t), A(t)]) \le \omega_1 \mathbf{v}\cdot[I_1(t), I_2(t), A(t)], \tag{2.7}$$

where "$\cdot$" denotes the vector dot product. Integrating the inequality (2.7) we have

$$0 \le \mathbf{v}\cdot[I_1(t), I_2(t), A(t)] \le \mathbf{v}\cdot[I_1(t), I_2(t), A(t)]e^{\omega_1(t-t_1)}, \quad \text{for } t_0 \le t_1 \le t.$$

Clearly, it follows that $\mathbf{v}\cdot[I_1(t), I_2(t), A(t)] \to 0$ as $t \to \infty$, since $\omega_1 < 0$. Since the components of $\mathbf{v}$ are all positive it means that

$$[I_1(t), I_2(t), A(t)] \to (0,0,0) \quad \text{as } t \to \infty.$$

By Lemma 2.2. we can choose two sequences $t_n \to \infty$ and $l_n \to \infty$ as $n \to \infty$, such that $S(t_n) \to S^\infty$, $\frac{dS(t_n)}{dt} \to 0$, $S(l_n) \to S_\infty$, $\frac{dS(l_n)}{dt} \to 0$. Since $(I_1(t), I_2(t), A(t)) \to (0,0,0)$ as $t \to \infty$ we obtain from the first equation of system (2.1) that

$$\mu b - \mu \limsup_{t\to\infty} S(t) = \mu b - \mu \liminf_{t\to\infty} S(t) = 0.$$

We thus have

$$\lim_{t\to\infty} S(t) = b.$$

Using Lemma 2.2. the disease free equilibrium point $\mathcal{E}_0$ is globally asymptotically stable if $\mathcal{R}_0 < 1$. This completes the proof. $\square$

2.3. *Existence and uniqueness of the endemic equilibrium*

The standard way to compute equilibria, is by setting the left hand side of system (2.1) to zero. From the equations of (2.1) we have

$$I_2^* = \frac{\sigma}{\mu + \rho} I_1^*, \tag{2.8}$$

$$A^* = \frac{\sigma\rho}{(\mu + \rho)(\mu + \delta)} I_1^* \tag{2.9}$$

$$\lambda^*(\mathbf{I}, A) = \frac{\beta c \Gamma I_1^*}{1 + \alpha \Gamma I_1^*}, \tag{2.10}$$

where

$$\Gamma = 1 + \frac{\eta_1 \sigma}{\rho + \mu} + \frac{\eta_2 \rho \sigma (1 - q)}{(\rho + \mu)(\delta + \mu)}.$$

Substituting $\lambda^*(\mathbf{I}, A)$ in the second equation of (2.1), gives $I_1^* = 0$, corresponding to the disease free equilibrium $(b, 0, 0, 0)$ and

$$S^* = \frac{(\sigma + \mu)(1 + \alpha \Gamma I_1^*)}{\beta c \Gamma}. \tag{2.11}$$

Adding the first two equations of (2.1), gives

$$\mu b - \mu S^* - (\sigma + \mu) I_1^* = 0$$

and substituting for S^* in the above we get an in equation in I_1^*. From this equation we find I_1^* as

$$I_1^* = \frac{\mu(\mathcal{R}_0 - 1)}{\alpha\mu + \beta c \Gamma},$$

If $\mathcal{R}_0 > 1$ then only one positive solution of I_1^* exists and no other positive solution exists for $\mathcal{R}_0 < 1$. A back substitution allows us to find I_2^*, A^* and S^* given by (2.8), (2.9) and (2.11) respectively, so that a unique endemic equilibrium point exists and is given by $\mathcal{E}_1 = (S^*, I_1^*, I_2^*, A^*) \in \mathcal{G}$, where

$$S^* = \frac{(\sigma + \mu)(\alpha\mu + \beta c \Gamma + \alpha\mu(\mathcal{R}_0 - 1))}{\beta c \Gamma (\alpha\mu + \beta c \Gamma)}, \tag{2.12}$$

$$I_1^* = \frac{\mu(\mathcal{R}_0 - 1)}{\alpha\mu + \beta c \Gamma}, \tag{2.13}$$

$$I_2^* = \frac{\sigma\mu(\mathcal{R}_0 - 1)}{(\rho + \mu)(\alpha\mu + \beta c \Gamma)}, \tag{2.14}$$

$$A^* = \frac{\rho\sigma\mu(\mathcal{R}_0 - 1)}{(\rho + \mu)(\delta + \mu)(\alpha\mu + \beta c \Gamma)}. \tag{2.15}$$

Global stability of E_0 in $\mathcal{G}$ excludes the existence of any other equilibrium other than $\mathcal{E}_0$ when ever $\mathcal{R}_0 < 1$. The endemic equilibrium is thus restricted to $\mathcal{R}_0 > 1$. We thus have the following theorem on the existence of the endemic equilibrium

Theorem 2.3. *If* $\mathcal{R}_0 > 1$, *the system* (2.1) *has a unique endemic equilibrium given* $\mathcal{E}_1$ *in* $\mathcal{G}$.

Remark. If $\mathcal{R}_0 = 1$, the endemic equilibrium $\mathcal{E}_1$ reduces to the disease-free equilibrium $\mathcal{E}_0$. The potential of the spread of infection in a population depends on the reproduction number. It is defined as the number of secondary infections produced by an index case in a population that is totally susceptible and at equilibrium.

2.4. *Local stability of the endemic equilibrium*

The local stability of the endemic equilibrium point $\mathcal{E}_1$, is determined by considering the signs of the eigenvalues of the Jacobian matrix of system (2.1) evaluated at the endemic equilibrium point. The Jacobian matrix of system (2.1) at $\mathcal{E}_1$ is given by

$$J^{\mathcal{E}_1} = \begin{pmatrix} -(\mu+\phi_1) & -\phi_2 & -\eta_1\phi_2 & -(1-q)\eta_2\phi_2 \\ \phi_1 & \phi_2-(\mu+\sigma) & \eta_1\phi_2 & (1-q)\eta_2\phi_2 \\ 0 & \sigma & -(\mu+\rho) & 0 \\ 0 & 0 & \rho & -(\mu+\delta) \end{pmatrix},$$

where $\phi_1 = \frac{c\beta g(\mathbf{I}^*,A^*)}{\psi(\mathbf{I}^*,A^*)}$ and $\phi_2 = \frac{c\beta S^*}{\psi(\mathbf{I}^*,A^*)}\left(1 - \frac{\alpha g(\mathbf{I}^*,A^*)}{\psi(\mathbf{I}^*,A^*)}\right)$.

Also note that $0 < \frac{\alpha g(\mathbf{I}^*,A^*)}{\psi(\mathbf{I}^*,A^*)} < 1$, implying that ϕ_1, ϕ_2 are both positive.

The characteristic equation of $J^{\mathcal{E}_1}$ is given by

$$P(\vartheta) = \vartheta^4 + \bar{a}_3\vartheta^3 + \bar{a}_2\vartheta^2 + \bar{a}_1\vartheta + \bar{a}_0 = 0, \tag{2.16}$$

where

$$\begin{aligned}
\bar{a}_3 &= (\mu+\sigma) + (\mu+\rho) + (2\mu+\delta) + (\phi_1-\phi_2),\\
\bar{a}_2 &= (\phi_1-\phi_2)(3\mu+\delta+\rho) + \sigma(\phi_1-\phi_2\eta_1) + (2\mu+\rho+\sigma)(3\mu+\delta) + \mu\delta + \rho\sigma,\\
\bar{a}_1 &= (\phi_1-\phi_2)[(2\mu+\delta)(\mu+\rho) + \mu(\mu+\delta)] + \sigma(2\mu+\delta)(\phi_1-\phi_2\eta_1)\\
&\quad + \rho\sigma\{\phi_1-\phi_2(1-q)\eta_2\} + \kappa\\
\bar{a}_0 &= (\mu+\sigma)(\mu+\delta)(\mu+\rho)(\mu+\phi_1)\left[\frac{\mu\phi_2\Gamma}{(\mu+\sigma)(\mu+\phi_1)} - 1\right],
\end{aligned}$$

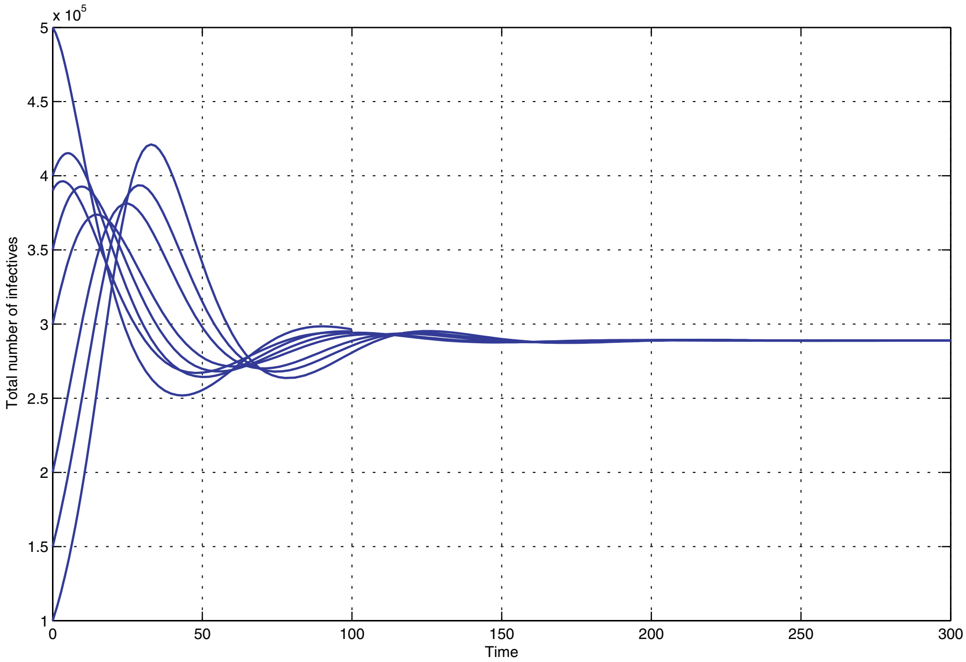
where $\kappa = \mu^2[3(\rho+\sigma+\delta)] + 2\mu\rho(\sigma+\delta) + \delta\sigma(2\mu+\rho)$. Some tedious algebraic manipulations give $\mathcal{R}_0 = \frac{\mu\phi_2\Gamma}{(\mu+\sigma)(\mu+\phi_1)}$. This means that $\bar{a}_0 > 0$ for $\mathcal{R}_0 > 1$.

Due to the mathematical intractability of the calculations involved in trying to establish the Routh-Hurwitz conditions for the stability of $\mathcal{E}_1$, we simply state the conditions under which system (2.1) has a locally asymptotically stable endemic equilibrium point.

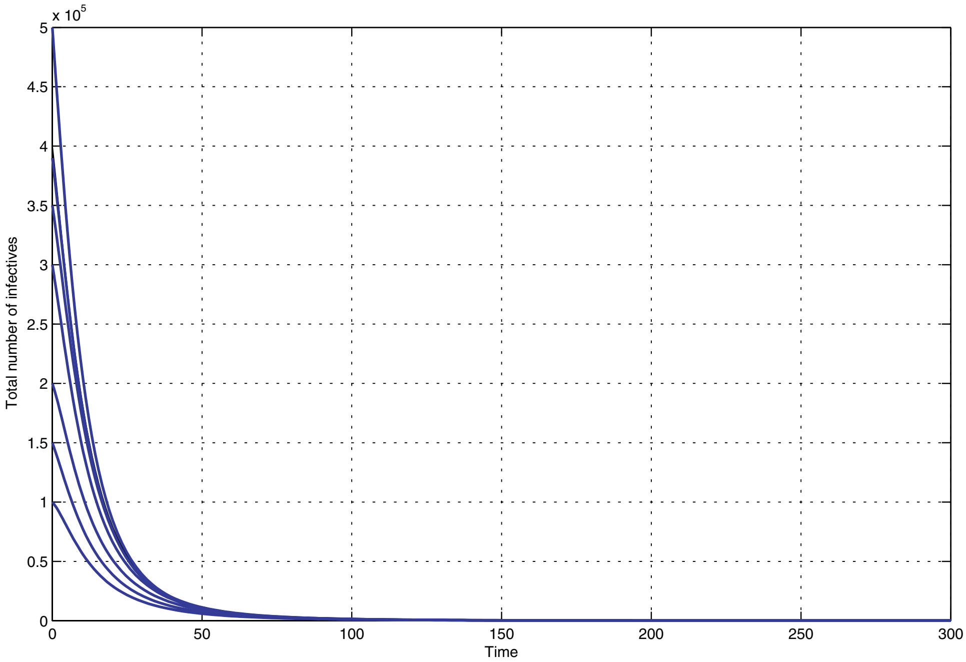
If $\bar{a}_3 > 0, \bar{a}_3\bar{a}_2 - \bar{a}_1 > 0$ and $\bar{a}_1[\bar{a}_3\bar{a}_2 - \bar{a}_1] - \bar{a}_3^2\bar{a}_0 > 0$, then the polynomial (2.16) has roots with negative real parts. We thus have the following result:

Theorem 2.4. *If $\mathcal{R}_0 > 1$, the endemic equilibrium $\mathcal{E}_1$ is locally asymptotically stable.*

Numerical simulations of the system (2.1) for specified parameter values indicate that irrespective of the initial conditions, the epidemic will always settle to an endemic steady state whenever $\mathcal{R}_0 > 1$, see Fig. 1.



(a)



(b)

Fig. 1. shows the variation of the total number of infectives for (a) $\mathcal{R}_0 > 1$ and (b) $\mathcal{R}_0 < 1$.

3. Numerical Simulations

To study the behavior of system (2.1) numerically, a fourth order Runge-Kutta scheme is used. We consider a hypothetical population of less than one million sexually active individuals. This would suit countries with small populations such as Botswana, Lesotho and Swaziland in sub-Sahara Africa. Vardavas and Blower,[31] reported that Botswana has approximately 730, 000 sexually active adults. An annual increase of 49,300 individuals was assumed in the same paper. In this paper $\mu = (0.02, 0.03)$ corresponding to an average sexual life of between 33 and 50 years, see also Refs. 1, 3, 25, 31 and 32. The susceptible population is assumed to increase at a rate of 0.05 per year, and this is equivalent to 40,000 individuals per year. The value of b is chosen in such a way that the rate at which susceptibles are recruited is 0.05 per year. The rate at which individuals exit the symptomatic HIV infected class varies; Naresh *et al.*[32] reports that individuals spend on average 2.5 years in the class corresponding to a rate of 0.4 per year; in Cai *et al.*[1] a rate of 0.02 per year is assumed. We take ρ to be between 0.05 and 0.06 per year. In Vardavas and Blower,[31] individuals spend an average of ten years before becoming symptomatic, corresponding to a rate of 0.1 per year. In this paper we consider an average value of between 0.14 and 0.18 per year for σ. This corresponds to an average duration of five to eight years before an individual becomes symptomatic. The relative infectivity estimates of individuals who are symptomatic and those with AIDS to asymptomatic individuals, represented in the model by η_1 and η_2 are not available. In Cai *et al.*,[1] symptomatic individuals are assumed to have a lower rate of infection when compared to the asymptomatic individuals. In this model we however assume that symptomatic individuals have a higher viral load and are thus more infectious, since viral load affects infection rate. The higher the viral load the higher the likelihood of infection.[33] We thus consider $1 \leq \eta_1, \eta_2 \leq 2$ in our simulations. One can however argue that the viral load argument can be overridden by the long duration of infectivity of the asymptomatic cases. However, our simulations show that similar results are obtained when $0 < \eta_1, \eta_2 < 1$, but at lower disease prevalence due to reduced infectivity of the symptomatic class. The probability of infection β varies considerably. In, Tuckwella *et al.*,[34] the probability was shown to be as low as 4×10^{-8} and as high as 1, especially in the presence of an intervention, with the highest probability occurring during the acute phase of infection. The death rate due to the disease is chosen to be 0.33 per year following.[35] Table 1 summarizes all the parameters used in the simulations.

We begin by considering numerical simulations that verify the theorems on the stability of the disease free and endemic equilibrium by considering the variation of each individual population for a given set of parameter values. Figure 2 illustrates the variation of S, I_1, I_2 and A with time whenever $\mathcal{R}_0 < 1$. For the case $\mathcal{R}_0 > 1$ we have the illustration of Fig. 3.

Table 1. Model parameters and their interpretations.

Parameter	Symbol	Value	Source
Recruitment rate	μb	0.05	Estimate
Proportion of withdrawals by AIDS cases	q	$0 \leq q \leq 1$	Estimate
Rate of becoming symptomatic	σ	$0.14 \leq \sigma \leq 0.18$	Estimate
Natural death rate	μ	$0.02 \leq \mu \leq 0.03$	Refs. 1, 3, 25, 31, 32
Exposure rate to media campaigns	α	$0 < \alpha$	Estimate
Rate of developing AIDS	ρ	$0.05 \leq \rho \leq 0.06$	Estimate
Probability of transmission	β	$0. \leq \beta \leq 1$	Ref. 34
Partner acquisition rate	c	$1, 1.5$	Ref. 3
Enhancement factors	(η_1, η_2)	$1 \leq \eta_1, \eta_2 \leq 2$	Ref. 1
Disease-induced death rate	δ	0.33	Ref. 35

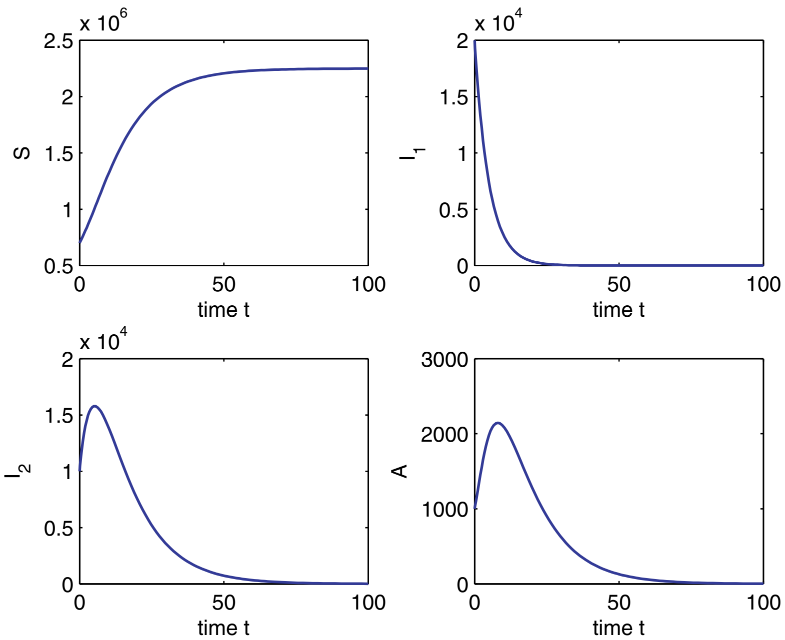


Fig. 2. Simulations for system (2.1) showing plots of the dynamics of all individual populations for the following parameter values $\mu = 0.02,\ \alpha = 1,\ \rho = 0.06,\ \beta = 0.00000002,\ \sigma = 0.18,\ \eta_1 = 1.6,\ \eta_2 = 1.8,\ q = 0.1,\ \delta = 0.33,\ c = 1$.

3.1. *Effects of increasing effectiveness of information campaigns*

The numerical simulations for assessing the effects of increasing the effectiveness of information campaigns are given in Fig. 4. In the simulations, we hypothetically assume that information campaigns start ten years after the start of the epidemic. The value of α is increased by 5%, 10%, 15% and 20% of the initially assigned value of α. The results in Fig. 4 illustrate that an increase in the effectiveness of

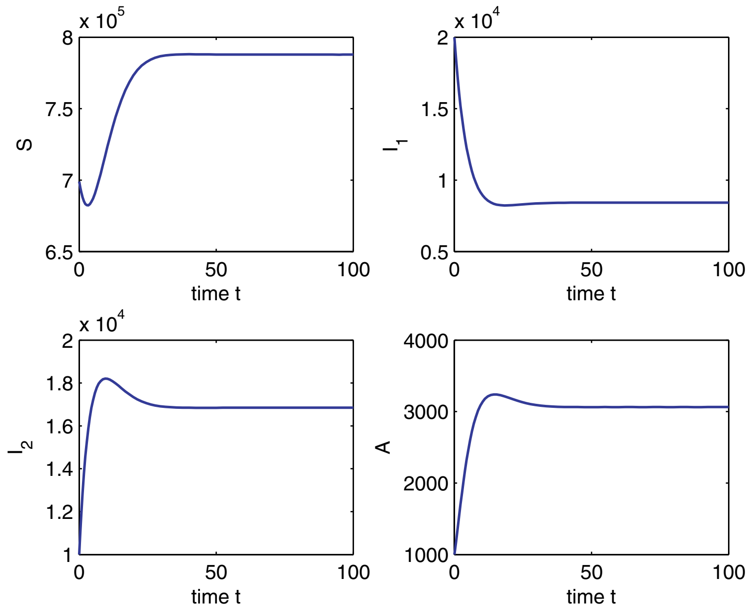


Fig. 3. Simulation results showing the dynamics of all individual populations for the following parameter values $\mu = 0.03$, $\alpha = 0.0006$, $\rho = 0.06$, $\beta = 0.0000009$, $\sigma = 0.18$, $\eta_1 = 1.2$, $\eta_2 = 1.8$, $q = 0.1$, $\delta = 0.33$, $c = 1.5$.

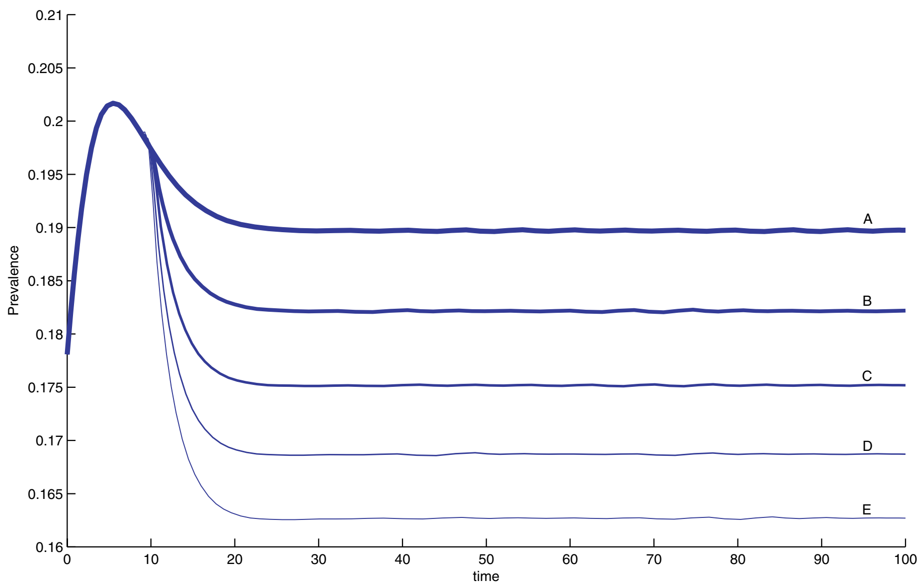


Fig. 4. Simulation results showing the change in prevalence due to increasing α for the following parameter values $\mu = 0.03$, $\rho = 0.05$, $\beta = 0.00000085$, $\sigma = 0.14$, $\eta_1 = 1.2$, $\eta_2 = 1.8$, $q = 0.1$, $\delta = 0.33$, $c = 1.5$. The scenarios A, B, C, D and E respectively represent changes in α as the values of α are increased by 5%, 10%, 15% and 20%, with A being the case for $\alpha = 0.000095$.

information campaigns results in a decrease in the prevalence of the infection. In the event that the effectiveness of information campaigns can be quantified, from the simulations, we observe that a 5% increase in the effectiveness of a campaign can lead to a reduction in the prevalence of around 0.7%. One can see the potential benefits of reducing the contact rate through increased information campaigns.

3.2. *Effects of withdrawing from sexual activities*

The effects of withdrawing from sexual activities, obtained by changing the values of q, were explored in Fig. 5. The numerical results show that an increase in the number of individuals with AIDS who withdraw from sexual activities is beneficial in reducing the spread of the epidemic as evidenced by the decline in the prevalence of the disease with increasing values of q. We note that an increase in the proportion q of 0.2 leads to a decline of approximately 0.3%.

3.3. *Effects of combining withdrawal from sexual activities and information campaigns*

Fig. 6 provides insights into the possible effects of combining withdrawal from sexual activities by individuals with AIDS and increasing information campaigns.

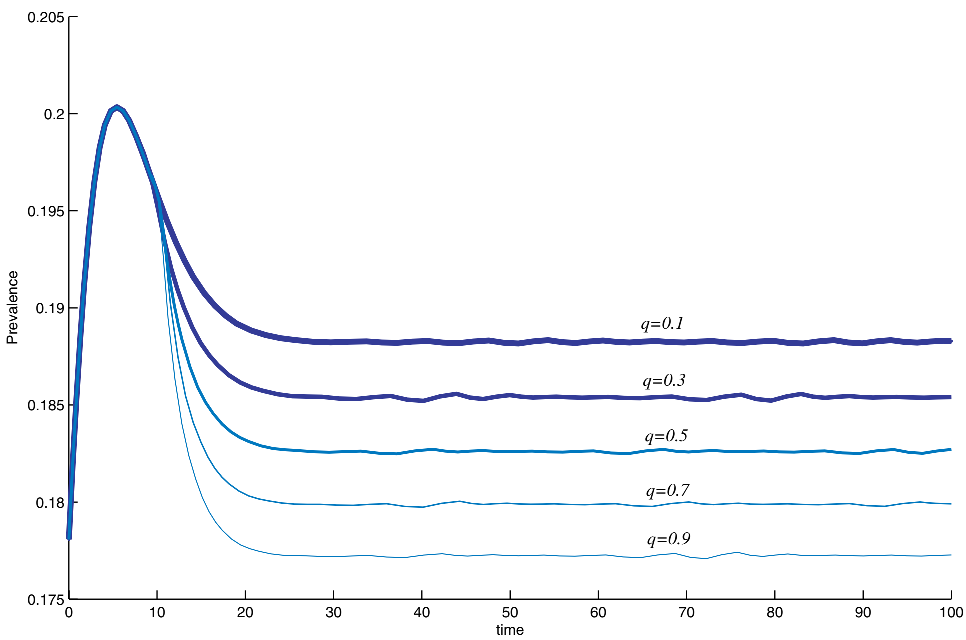


Fig. 5. Simulation results showing the change in prevalence due to increasing q for the following parameter values $\mu = 0.03$, $\rho = 0.05$, $\beta = 0.00000085$, $\alpha = 0.000095$, $\sigma = 0.14$, $\eta_1 = 1.2$, $\eta_2 = 1.8$, $\delta = 0.33$, $c = 1.5$.

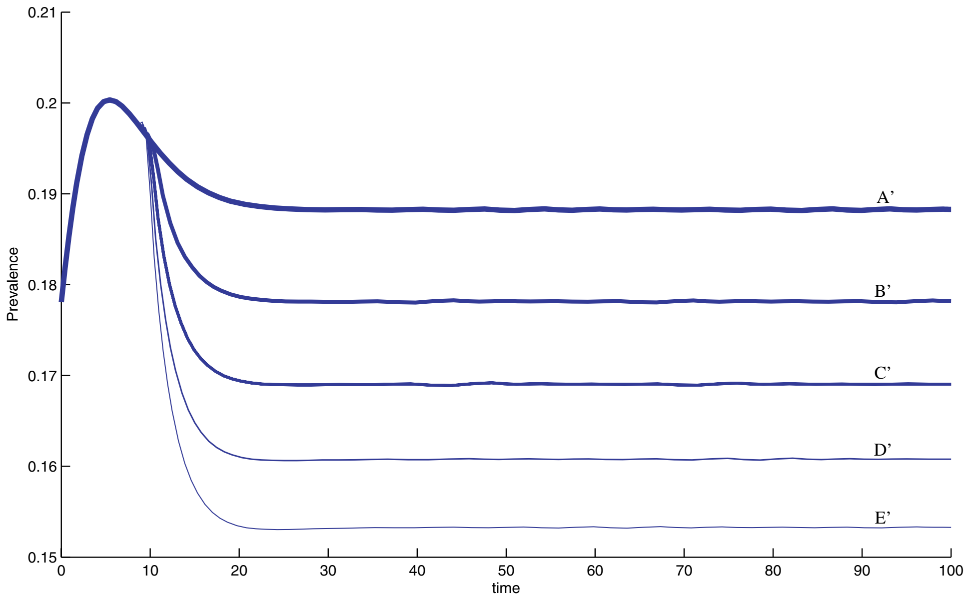


Fig. 6. Simulation results showing the change in prevalence due to increasing α and q for the following parameter values $\mu = 0.03$, $\rho = 0.05$, $\beta = 0.00000085$, $\sigma = 0.14$, $\eta_1 = 1.2$, $\eta_2 = 1.8$, $\delta = 0.33$, $c = 1.5$. The scenarios A', B', C', D' and E' respectively represent changes in α and q as the values of α are increased by 5%, 10%, 15% and 20%, with A being the case for $\alpha = 0.000095$ and q is as given in Fig. 5.

The values of α and q are varied as presented in the Figs. 4 and 5. The simulation results show that for each increase in the values of α and q, the prevalence fall by 1%. We thus conclude the importance of combining prevention strategies in the fight against HIV/AIDS, as also noted in Ref. 3.

4. Discussion

Weir *et al.*[36] acknowledged the need to do research that evaluates the effectiveness of prevention strategies. Prevention efforts such as school education, work place education and mass media campaigns also need evaluation. In this paper, a model for HIV/AIDS that incorporates HIV prevention using mass media campaigns and withdrawal of individuals with AIDS is considered. Reduction in disease transmission due to media campaigns is modeled by aa incidence function with saturation. The function can be classified as a 'Hill plot'[18] and has been used to model physiological and biochemical processes, see for instance.[37,38] Saturation can be used to model, what we will refer to as *prevention fatigue* in this paper. Prevention fatigue mirrors complacency. It is defined by a positive response to an intervention program, leading to a fast decline in the contact rate. This is then followed by a sustained slow decline of the contact rate. It remains one of the biggest threats to accelerated and sustainable HIV prevention efforts.[39] Prevention fatigue leads to a relapse of risky sexual behavior and this has been observed recently in some countries.[11,40,41]

The dynamical behavior of system (2.1) is presented in terms of the model reproduction number $\mathcal{R}_0$. If $\mathcal{R}_0 < 1$, the disease free equilibrium point is both locally and globally asymptotically stable. On the other hand if $\mathcal{R}_0 > 1$, the disease persists in the population and a unique endemic equilibrium point exists and is locally asymptotically stable. Conditions under which the disease persists in the population are also derived. The impact of media campaigns on HIV transmission is measured by the parameter α. Although the model reproduction number $\mathcal{R}_0$ does not explicitly depend on α, numerical simulation results show that an increase in α leads to a reduction in the prevalence of the disease. Similar conclusions can be made regarding the proportion q of individuals with AIDS who withdraw from sexual activities. Increasing q leads to a reduction in the prevalence of infection. The model can be used to quantify the role played by media campaigns and how they can possibly reduce the prevalence of the disease. Such a model can be useful in sub-Saharan Africa where information through the media is currently being used extensively in the fight against HIV. The lack of a proven effective vaccine to stop HIV transmission has lead to much of public policy putting an emphasis on information campaigns in order to reduce HIV-prevalence. As shown previously in Ref. 3, media campaigns alone may not be the ultimate strategy in the fight against HIV/AIDS. Inclusion of other intervention strategies is thus imperative.

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